

Quiz 1: DIC on Discrete Structures

20 marks, 50min

1 Sep 2023

Instructions:

- Attempt *all* questions. Write all answers and proofs carefully. If you are *making any assumptions or using results proved in class*, state them clearly.
- All sets considered below are general sets that can be infinite. Hence, you must not assume that they are finite or countable, unless clearly specified otherwise.
- In this course, one of our aims is to learn how to write good proofs, hence considerable weightage will be given to *clarity and completeness* of proofs.
- Do not copy or use any other unfair means. Offenders will be reported to the Disciplinary Action Committee.

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1. [2 marks] Prove that, if a, b are both prime numbers (> 1), and $a \neq b$, then $\log_b a$ is irrational.
 2. [2 marks] Answer True or False. Also give a 1-line justification for each.
 - (a) Every non-empty subset of integers has a smallest element.
 - (b) If a proposition is true, then its converse must also be true.
 3. [3 marks] Prove by induction that for all $n \in \mathbb{N}$, 3^n is odd.
 4. [4 marks] Prove or disprove: For any two sets A, B , there exists a surjection from A to B iff there exists an injection from B to A .
 5. Which of the following sets are countable? Justify with formal proof. You may assume that countable union of countable sets is countable.
 - (a) [4 marks] Set of all functions from $\mathbb{N}$ to $\mathbb{N}$.
 - (b) [5 marks] Set of all non-increasing functions from $\mathbb{N}$ to $\mathbb{N}$. A function $f : \mathbb{N} \rightarrow \mathbb{N}$ is said to be non-increasing if for all $x, y \in \mathbb{N}$, if $x \leq y$ then $f(x) \geq f(y)$.